

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 22: Integral Calculus

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1 Keywords

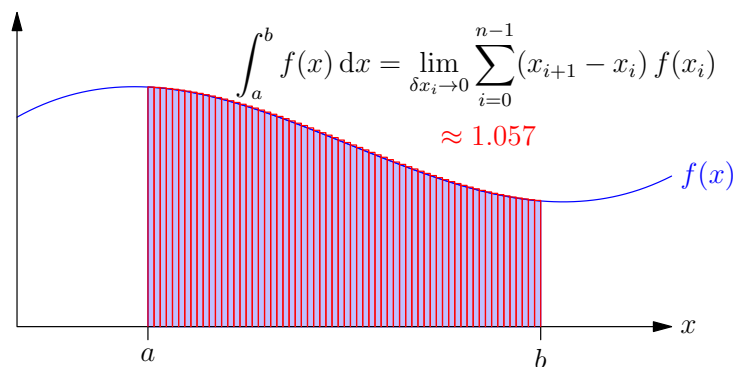
- Riemann integral, fundamental theorem of calculus, integration by parts, substitution, Jacobians, generating functions, gamma function, Gaussian integrals

2 Main Points

2.1 The Riemann Integral

- An integral is the area under a curve. Divide $[a, b]$ into strips at points $a = x_0 < x_1 < \dots < x_n = b$ and add up the areas of the rectangles:

$$\int_a^b f(x) dx = \lim_{\delta x_i \rightarrow 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) f(x_i)$$



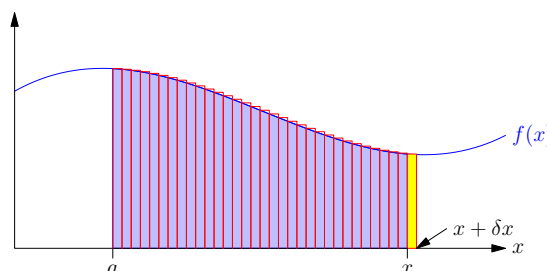
- Integration is linear: splitting the sum,

$$\int_a^b (r f(x) + s g(x)) dx = r \int_a^b f(x) dx + s \int_a^b g(x) dx$$

2.2 The Fundamental Theorem of Calculus

- Let $I(a, x) = \int_a^x f(z) dz$. Extending the upper limit by δx adds a thin strip of area $\delta x f(x)$, so

$$\frac{dI(a, x)}{dx} = \lim_{\delta x \rightarrow 0} \frac{I(a, x + \delta x) - I(a, x)}{\delta x} = f(x)$$



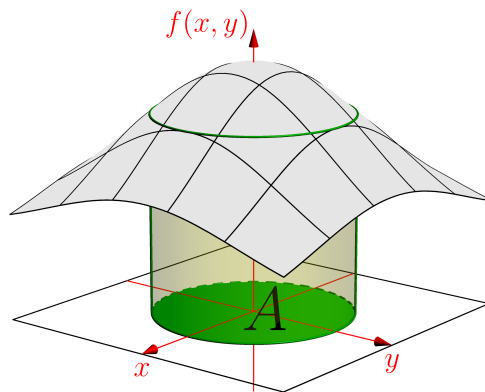
- Conversely, integrating a derivative, the Riemann sum telescopes:

$$\begin{aligned}\int_a^b \frac{df(x)}{dx} dx &= \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} (x_{i+1} - x_i) \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} (f(x_{i+1}) - f(x_i)) \\ &= f(x_n) - f(x_0) = f(b) - f(a)\end{aligned}$$

- So integration is an *anti-derivative*: it undoes differentiation

2.3 Indefinite and Multiple Integrals

- An *indefinite integral* is a function $F(x) = \int f(x) dx$ with $F'(x) = f(x)$. Since $F(x)$, $F(x) + 1$, $F(x) + \pi, \dots$ all have the same derivative, F is only defined up to an additive constant
- Definite integrals follow from it: $\int_a^b f(x) dx = F(b) - F(a)$
- For functions of several variables we integrate over regions, e.g. $\iint_A f(x, y) dx dy$. We usually write a single integral sign, $\int f(x) dx$, for $\int \dots \int f(x_1, \dots, x_n) dx_1 \dots dx_n$



2.4 Performing Integrals

- The basic method is to recognise an anti-derivative. Since $dx^n/dx = n x^{n-1}$,

$$\int x^{n-1} dx = \frac{x^n}{n} + c \quad (n \neq 0), \quad \int_a^b x^{n-1} dx = \frac{b^n - a^n}{n}$$

- The product and chain rules let us differentiate almost anything, but there are no general rules for $\int f(x) g(x) dx$ or $\int f(g(x)) dx$. Unlike differentiation there is no small parameter to expand in. *In general integration is hard*

2.5 Integration by Parts

- Integrating the product rule $d(fg)/dx = f'g + fg'$ from a to b and rearranging,

$$\int_a^b f(x) \frac{dg(x)}{dx} dx = \left[f(x) g(x) \right]_a^b - \int_a^b \frac{df(x)}{dx} g(x) dx$$

Whether this helps depends on whether the new integral is easier

- **Example: the factorial function.** Let $\Pi(z) = \int_0^\infty x^z e^{-x} dx$. Taking $g(x) = -e^{-x}$,

$$\Pi(z) = \left[-x^z e^{-x} \right]_0^\infty + \int_0^\infty z x^{z-1} e^{-x} dx = z \Pi(z-1)$$

since the boundary term vanishes for $z > 0$

- $\Pi(0) = \int_0^\infty e^{-x} dx = \left[-e^{-x} \right]_0^\infty = 1$, so for integers $\Pi(n) = n(n-1) \dots 1 = n!$

2.6 Substitution

- Changing variable from x to u , with $x = x(u)$, the Riemann sum becomes

$$\int_a^b f(x) dx = \int_{u(a)}^{u(b)} f(x(u)) \frac{dx(u)}{du} du$$

where $u(x)$ is the inverse of $x(u)$

- Example.** For $I(n) = \int_0^\infty x^n e^{-x^2/2} dx$, put $u = x^2/2$, so $x = \sqrt{2u}$, $dx/du = 1/\sqrt{2u}$, and the limits stay 0 and ∞ :

$$I(n) = \int_0^\infty (2u)^{n/2} e^{-u} \frac{du}{\sqrt{2u}} = 2^{\frac{n-1}{2}} \int_0^\infty u^{\frac{n-1}{2}} e^{-u} du = 2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2} + 1\right)$$

- So $I(1) = 1$, $I(3) = 2 \times 1! = 2$ and $I(5) = 2^2 \times 2! = 8$. Even n need Γ at half integers, e.g. $I(0) = \Gamma(1/2)/\sqrt{2}$, and $I(2) = \sqrt{2} \Gamma(3/2) = \Gamma(1/2)/\sqrt{2}$ using $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2)$

2.7 Changing Variables in Many Dimensions

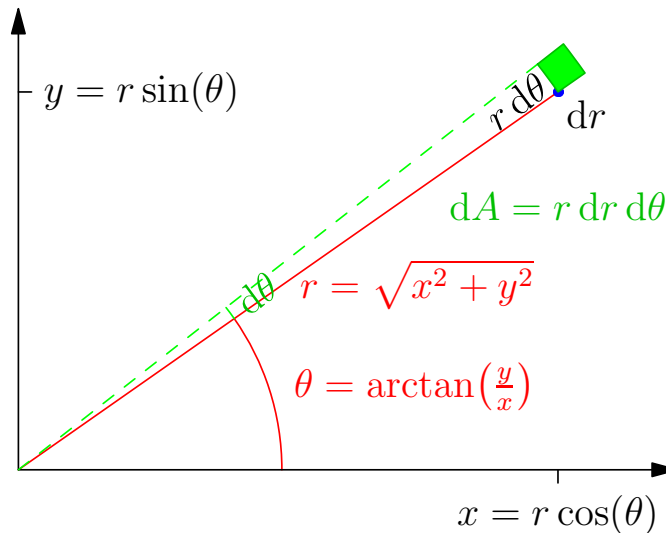
- Changing variables $x \rightarrow u$, volume elements are rescaled by the determinant of the Jacobian:

$$\int f(x) dx = \int f(x(u)) |\det(\mathbf{J})| du, \quad J_{ij} = \frac{\partial x_i}{\partial u_j}$$

- Polar coordinates.** With $x = r \cos(\theta)$ and $y = r \sin(\theta)$,

$$|\det(\mathbf{J})| = \left| \det \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix} \right| = r (\cos^2(\theta) + \sin^2(\theta)) = r$$

so $dx dy = r dr d\theta$: a small patch at radius r has sides dr and $r d\theta$



2.8 Differentiating Through the Integral

- Sometimes we can do an integral by differentiating a related, easier one. The moments of a random variable are $M_n = \mathbb{E}[X^n] = \int x^n f_X(x) dx$
- Define the *moment generating function*

$$Z(\ell) = \mathbb{E}[e^{\ell X}] = \int_{-\infty}^{\infty} e^{\ell x} f_X(x) dx$$

- Differentiating n times under the integral and setting $\ell = 0$,

$$\left. \frac{d^n Z(\ell)}{d\ell^n} \right|_{\ell=0} = \int_{-\infty}^{\infty} x^n f_X(x) dx = M_n$$

2.9 The Cumulant Generating Function

- Expanding $e^{\ell x} = 1 + \ell x + \frac{1}{2}\ell^2 x^2 + \dots$ inside the integral,

$$Z(\ell) = 1 + \ell M_1 + \frac{1}{2}\ell^2 M_2 + \frac{1}{3!}\ell^3 M_3 + \dots$$

- Taking the logarithm, using $\log(1 + \epsilon) = \epsilon - \frac{1}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 - \dots$, gives the *cumulant generating function*

$$G(\ell) = \log(Z(\ell)) = \ell M_1 + \frac{1}{2}\ell^2 (M_2 - M_1^2) + \frac{1}{3!}\ell^3 (M_3 - 3 M_2 M_1 + 2 M_1^3) + \dots$$

- The *cumulants* are $\kappa_n = G^{(n)}(0)$: κ_1 is the mean, κ_2 the variance and κ_3 is related to the skewness

2.10 Special Functions

- Some integrals have no closed form in terms of elementary functions. $\Pi(z)$ is one: $\Pi(n) = n!$ for integers, but for general z it is a new function in its own right
- History has left us with the *gamma function* instead,

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx = \Pi(z-1)$$

- Other special functions defined by integrals include the Bessel, Airy, hypergeometric, elliptic and error functions
- Integration is sometimes easier in the complex plane (largely due to Cauchy), and there is an algorithm (the Risch algorithm) that decides whether many functions have elementary integrals. It is complicated enough that it is only available inside symbolic packages such as Mathematica. Both are beyond this course

2.11 Gaussian Integrals

- Integrals involving e^{-x^2} appear whenever we work with normal distributions. The good news is that they can all be done; the bad news is that they take some work
- The Gaussian integral.** $I_1 = \int_{-\infty}^\infty e^{-x^2/2} dx$ has no elementary anti-derivative, but its square does yield:

$$I_1^2 = \int_{-\infty}^\infty \int_{-\infty}^\infty e^{-(x^2+y^2)/2} dx dy = \int_0^{2\pi} d\theta \int_0^\infty r e^{-r^2/2} dr$$

changing to polar coordinates. Substituting $u = r^2/2$, $du = r dr$,

$$I_1^2 = 2\pi \int_0^\infty e^{-u} du = 2\pi \implies I_1 = \sqrt{2\pi}$$

- Since $I_1 = 2 I(0) = \sqrt{2} \Pi(-1/2)$, this also shows $\Gamma(1/2) = \Pi(-1/2) = \sqrt{\pi}$
- Normal distribution.** Substituting $z = (x - \mu)/\sigma$,

$$\int_{-\infty}^\infty e^{-(x-\mu)^2/(2\sigma^2)} dx = \sigma I_1 = \sqrt{2\pi} \sigma$$

which is why the normal density is

$$\mathcal{N}(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi} \sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

2.12 Multivariate Gaussian Integrals

- In n dimensions the exponential of a sum is a product of exponentials, so the integral factorises:

$$\int e^{-\frac{1}{2}\|\mathbf{x}\|_2^2} d\mathbf{x} = \prod_{i=1}^n \int_{-\infty}^{\infty} e^{-x_i^2/2} dx_i = (2\pi)^{n/2}$$

- For a general covariance matrix Ξ , consider

$$I_4 = \int e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Xi^{-1}(\mathbf{x}-\boldsymbol{\mu})} d\mathbf{x}$$

- Diagonalise $\Xi = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$, with $\mathbf{V}$ orthogonal, so $\Xi^{-1} = \mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^\top$. Change variables to $\mathbf{y} = \mathbf{V}^\top(\mathbf{x} - \boldsymbol{\mu})$, i.e. $\mathbf{x} = \mathbf{V}\mathbf{y} + \boldsymbol{\mu}$. The Jacobian is $J_{ij} = \partial x_i / \partial y_j = V_{ij}$, and $|\det(\mathbf{V})| = 1$ for an orthogonal matrix, so

$$I_4 = \int e^{-\frac{1}{2}\mathbf{y}^\top \boldsymbol{\Lambda}^{-1}\mathbf{y}} d\mathbf{y} = \prod_{i=1}^n \int_{-\infty}^{\infty} e^{-y_i^2/(2\lambda_i)} dy_i = \prod_{i=1}^n \sqrt{2\pi \lambda_i}$$

- Now $\det(\Xi) = \det(\mathbf{V}) \det(\boldsymbol{\Lambda}) \det(\mathbf{V}^\top) = \prod_i \lambda_i$, and $\det(c\mathbf{M}) = c^n \det(\mathbf{M})$ for an $n \times n$ matrix, so

$$I_4 = (2\pi)^{n/2} \sqrt{\det(\Xi)} = \sqrt{\det(2\pi \Xi)}$$

- The multivariate normal density, which must integrate to 1, is therefore

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \Xi) = \frac{1}{\sqrt{\det(2\pi \Xi)}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Xi^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

- Normal distributions almost always lead to integrals that can be done in closed form, although often with some effort. Being comfortable with integration is a genuine super-power

3 Exercises

3.1 Integration by parts

- Evaluate $\int_0^1 x e^x dx$

3.2 Substitution

- Evaluate $\int_0^\infty x e^{-x^2} dx$

3.3 Mean absolute value of a normal

- For a standard normal Z , use $I(1)$ to find $\mathbb{E}[|Z|]$

3.4 The normal moment generating function

- Show that for $X \sim \mathcal{N}(\mu, \sigma^2)$ the moment generating function is $Z(\ell) = e^{\mu\ell + \sigma^2\ell^2/2}$, by completing the square in the exponent. What does this say about the cumulants of a normal distribution?

4 Answers

4.1 Integration by parts

- Take $f(x) = x$ and $g(x) = e^x$:

$$\int_0^1 x e^x dx = \left[x e^x \right]_0^1 - \int_0^1 e^x dx = e - (e - 1) = 1$$

4.2 Substitution

- With $u = x^2$, $du = 2x dx$:

$$\int_0^\infty x e^{-x^2} dx = \frac{1}{2} \int_0^\infty e^{-u} du = \frac{1}{2}$$

4.3 Mean absolute value of a normal

- By symmetry

$$\mathbb{E}[|Z|] = \frac{2}{\sqrt{2\pi}} \int_0^\infty z e^{-z^2/2} dz = \frac{2 I(1)}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}} \approx 0.798$$

4.4 The normal moment generating function

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$$Z(\ell) = \frac{1}{\sqrt{2\pi}\sigma} \int e^{\ell x - (x-\mu)^2/(2\sigma^2)} dx$$

- Completing the square,

$$\ell x - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{(x-\mu-\sigma^2\ell)^2}{2\sigma^2} + \mu\ell + \frac{\sigma^2\ell^2}{2}$$

- The remaining integral is a normal density with mean $\mu + \sigma^2\ell$, which integrates to 1, so $Z(\ell) = e^{\mu\ell + \sigma^2\ell^2/2}$
- Hence $G(\ell) = \log Z(\ell) = \mu\ell + \frac{1}{2}\sigma^2\ell^2$ exactly: $\kappa_1 = \mu$, $\kappa_2 = \sigma^2$, and every higher cumulant is zero. The normal is the distribution whose cumulants stop after the second